

AI and The Dataverse® Project

Using AI to Benefit Research Data Community Members

The Dataverse® Project

Open Source Data Repository

137 installations, 38 countries

9 CoreTrustSeal repositories

Active Community

FAIR support

Metadata standards (e.g. **Croissant ML**)

Custom metadata

Citations, Embargoes, Provenance, other Features

Previewers

External tools

Interoperability/Application integrations



Harvard Dataverse Repository

Open to all researchers (not just Harvard's affiliates) from **any discipline** (generalist repository)

All file types and disciplines are welcome

All **data** must be **de-identified**

All datasets undergo ***curatorial review***

- 105K+ datasets
- 2.8+ million files
- 107+ million downloads

Dataverse Project Timeline

Dr. Gary King
Founder & Principal Investigator
Albert J. Weatherhead III University Professor
Harvard University
Institute for Quantitative Social Sciences


The 2023 Dataverse Community Meeting in Braga, Portugal

1997 ▸ 2006 ▸ 2007 ▸ 2008 ▸ 2015 ▸ 2016 ▸ 2023 ▸ 2024 ▸ 2025 ▸ 2026

1997
Project Begins

Focus on preserving & sharing Social Sciences data

2006
The Dataverse Network 1.0

Harvard Dataverse Repository established

2007
🎉 Announcing the Dataverse Network

2008
Odum Archive: First Dataverse outside Harvard

2015
Data Citations

🤝 First Open Dataverse Community Meeting

Dataverse 4.0

2016
FAIR Data

2023
🎉 100th Dataverse installation

Dataverse 6.0

Dataverse Re-architecture

2024
🤖 Ask the Data

🥐 Croissant

New Dataset Types (e.g., software, computational workflows)

137 Dataverse Installations

Dataverse 6.7

Large Data Services

Traditional Knowledge (TK) Labels

Dataverse MCP Server

Single-Page App Launch

2025
🚀 The Next Frontier

Locally FAIR Data

Trusted Data Labels

Dataverse Knowledge Graph

Dataverse and AI

- Investigations & development began around 2023 at Harvard and across the Dataverse Community
- Emphasis on generative AI, LLM, Retrieval-Augmented Generation (RAG), **human-in-the-loop**
- Many resources, tools, and projects are in production or underway
- **AI Promise**: Improve data findability; metadata quality; data reusability; tool development & integration; support data exploration & curation



Photo by Justin Ha on Unsplash

"A computer can never be held accountable, therefore a computer must never make a management decision".

IBM training manual, 1979

Who Benefits from Dataverse AI?

- Data curators
- Data depositors
- Data seekers
- Dataverse repository users
- Data reusers
- Metadata harvesters
- Repository managers
- Repository strategists
- Software developers
- System integrators



Image by Gerd Altmann at Pixabay

Dataverse AI-Related Resources, Tools & Projects

Resource, Tool or Project	Type	Improves	Status
AI Guide	Resource	User experience	Production
Ask the Data	Tool	Data reusability	Production
Ask Dataverse	Tool	User experience	Experimental
AutoSage	Tool	Metadata quality, User experience	Experimental
Croissant 🥐	Tool	Data findability	Production
Enhancing Dataset Metadata Project	Project (Tool)	Metadata quality, User experience	Development
GREI AI Taxonomy	Resource	Understanding of AI roles in RDM	Production
Research Data Metadata Knowledge Graph	Project (Tool)	Data findability, Data reusability	Development
Model Context Protocol (MCP) Server	Tool	Data findability, Data reusability	Production
Spam Detection & Workflow Automation	Project (Tool)	Repository data quality	Investigation
TurboCurator	Tool	Metadata quality, User experience	Production

Dataverse AI Guide

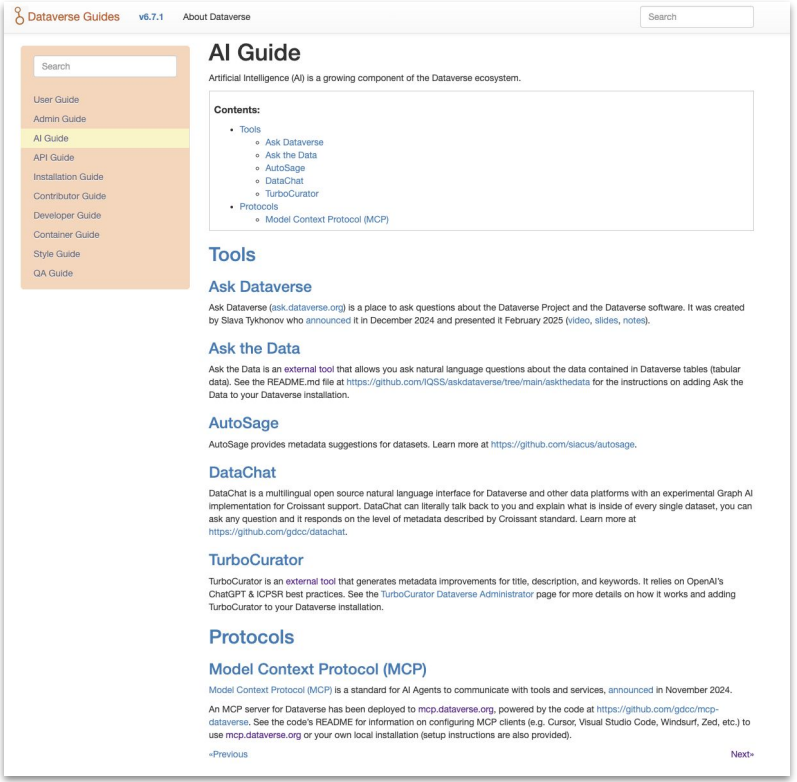
👏**Improves**: User experience
🤝**Who Benefits?** Dataverse repository users

About the Guide (link)

Goal: Provide Dataverse platform developers and users with access to supported AI tools and protocols

Currently Featured Tools

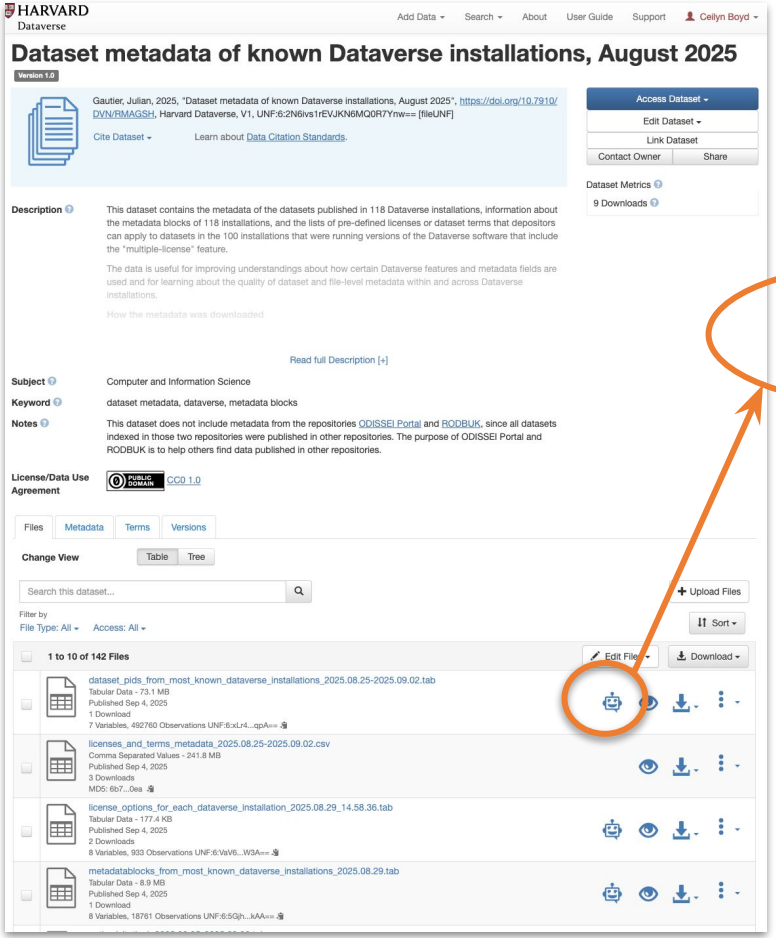
- Ask Dataverse
- Ask the Data
- AutoSage
- Data Chat
- TurboCurator
- Model Context Protocol (MCP) Server



Ask the Data 

👏**Improves**: Data reusability, User experience
🤝**Who Benefits?** Data depositors

- Chatbot for tabular data files based on OpenAI
- Shiny app, 200 lines of Python code
- Created as external Dataverse tool
- Transforms NL query into SQL, executes the query on the Dataverse database, translates results into natural language
- GitHub repo: https://github.com/IQSS/askdataverse/tree/main/askthedata



👏**Improves**: Data reusability, User experience

🤝**Who Benefits?** Data depositors

Tell me what you want to know

which installations have the most pids?

Answer please

The installations with the most PIDs are as follows: 1. **DANS Data Station Archaeology**: 163,458 PIDs 2. **Harvard Dataverse**: 106,285 PIDs 3. **Open Forest Data**: 79,774 PIDs 4. **JPL Open Repository**: 45,859 PIDs 5. **Borealis**: 22,960 PIDs These are the top installations based on the number of PIDs they have.

dataverse_installation_name	dataset_pid	dataset_pid_url	dataverse_collection_alias	dataverse_collection_name	dataverse_collection_type
Domus Dados	hdl:20.500.12682/rdp/136	https://hdl.handle.net/20.500.12682/rdp/136	unifesp	Domus Dados: repositório de dados de pesquisa da Universidade Federal de São Paulo	ORGANIZATIONS_INSTITUTIO
Domus Dados	hdl:20.500.12682/rdp/SEEKNF	https://hdl.handle.net/20.500.12682/rdp/SEEKNF	ggesepb	Greenhouse gas emissions from sugarcane ethanol production in Bolivia	RESEARCH_PROJECTS
Domus Dados	hdl:20.500.12682/rdp/9N2XCL	https://hdl.handle.net/20.500.12682/rdp/9N2XCL	ggesepb	Greenhouse gas emissions from sugarcane ethanol production in Bolivia	RESEARCH_PROJECTS
Domus Dados	hdl:20.500.12682/rdp/2QBYIR	https://hdl.handle.net/20.500.12682/rdp/2QBYIR	ggesepb	Greenhouse gas emissions from sugarcane ethanol production in Bolivia	RESEARCH_PROJECTS

Viewing rows 1 through 4 of 492401

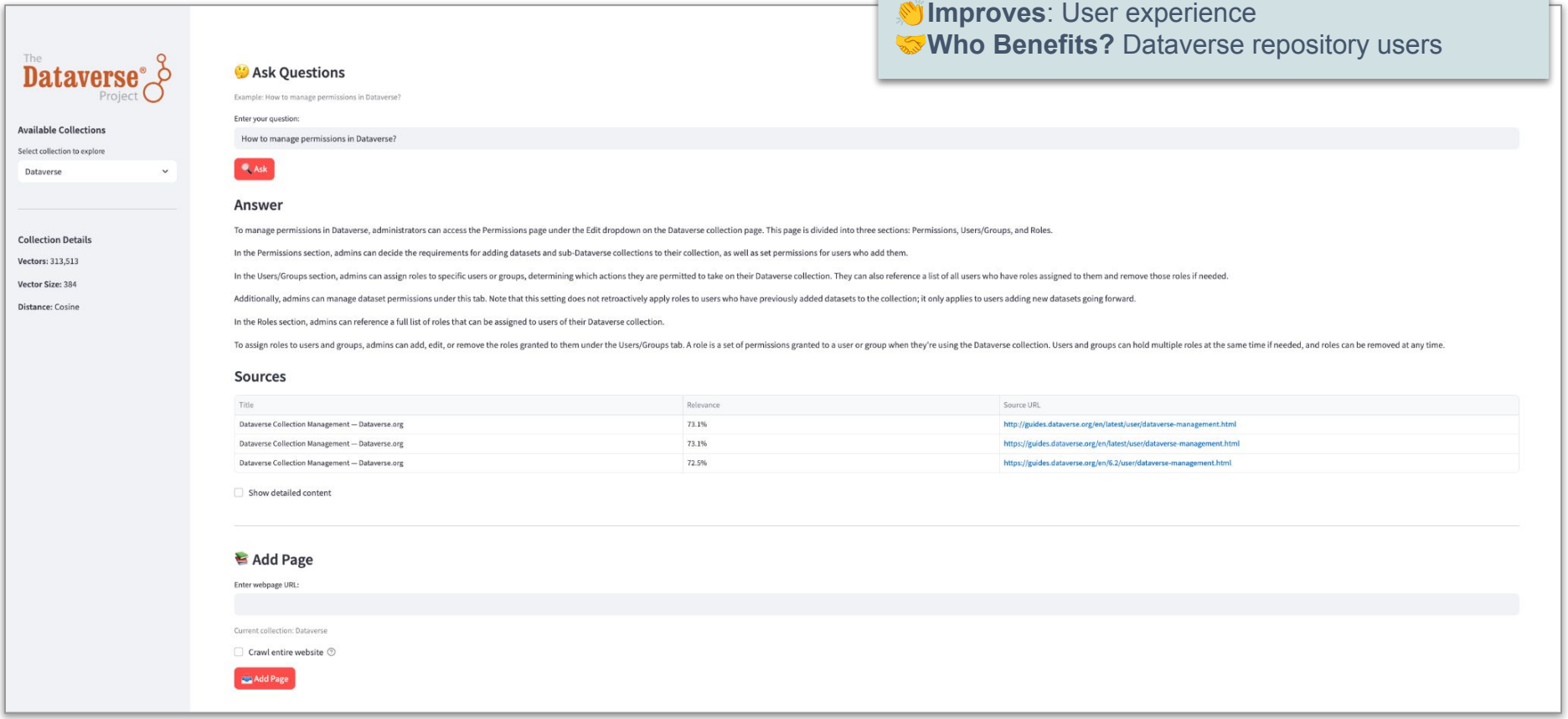
- Uses OpenAI
- Requires tabular data files
- NLP → SQL → NLP

Ask Dataverse

👏**Improves**: User experience
🤝**Who Benefits?** Dataverse repository users

- Website: ask.dataverse.org
- Created by Slava Tykhonov (DANS) in December 2024
- External tool designed as "a place to ask questions about the Dataverse Project and the Dataverse software"
- Trained on Dataverse documentation
- Ask Dataverse resources: video, slides, notes

👏**Improves**: User experience
🤝**Who Benefits?** Dataverse repository users



AutoSage AI Helper

👏**Improves**: Metadata quality, User experience
🤝**Who Benefits?** Data depositors, Data seekers

- R-shiny app
- Suggests additional dataset metadata (subjects & keywords) given a dataset DOI or its title and description
- Based on the LLAMA-2-7B and LLAMA-3.2-3B models
- Fine-tuned on the full Harvard Dataverse corpus
- GitHub repo: https://github.com/siacus/autosage?tab=readme-ov-file

- **First model** (LLAMA-2-7B) trained on 76,110 datasets to June 2024
- Model shows an accuracy of 94.6%
- Details and reference to scripts: https://arxiv.org/abs/2411.00890
- **Second model** (LLAMA-3.2-3B) trained on a subset of a clean and balanced training set (data balanced by subject category).
- Accuracy varies around 80%
- Paper is forthcoming

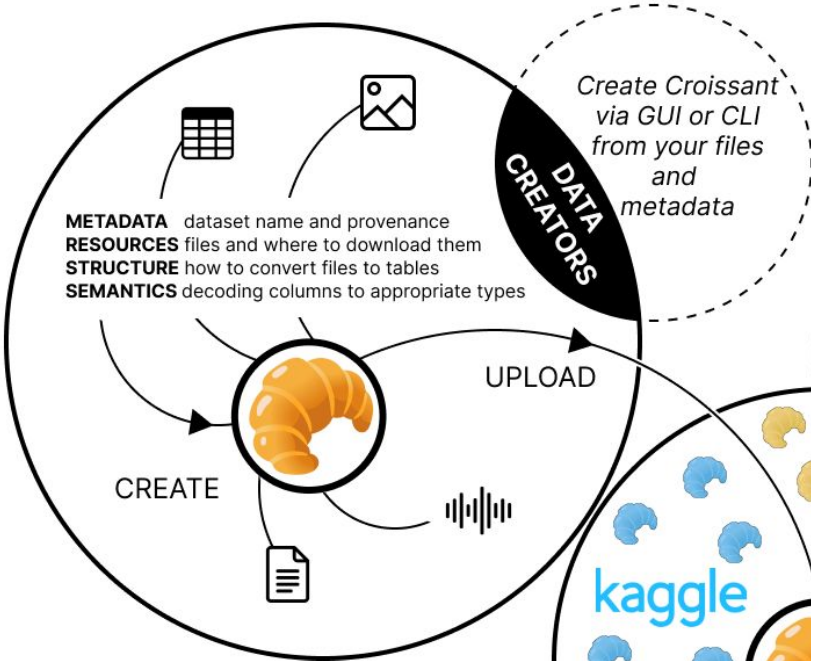


👏**Improves**: Metadata quality, User experience
🤝**Who Benefits?** Data depositors, Data seekers

Croissant 🥐

👏**Improves**: Data findability
🤝**Who Benefits?** Data seekers, Metadata harvesters

- "Croissant 🥐 is a high-level format for machine learning datasets that combines **metadata**, resource file descriptions, data structure, and default ML semantics into a single file; it works with existing datasets to make them easier to find, use, and support with tools."
- Built on the schema.org standard
- Dataverse supports Croissant metadata exporting
- Findable via Google Dataset Search



The GREI Data Repository AI Taxonomy

👋**Improves**: Understanding of AI roles in RDM
🤝**Who Benefits?** Product managers, Repository strategists

About Generalist Repository Ecosystem Initiative (GREI)

- Collaborating repositories



Goals

1. Improve researcher data sharing
2. Improve discoverability
3. Support reproducibility
4. Avoid duplication
5. Increase data reuse

About the GREI AI Taxonomy

- Classifies the distinct roles AI can play in data repository operations
- Provides a common language and reference point to foster productive discussions among stakeholders
- Provides framework to identify & address AI integration challenges in data repositories
- Chodacki, J., Hanhel, M., Iacus, S., Scherle, R., Olson, E., Pfeiffer, N., Holmes, K., & Hosseini, M. (2024). GREI Data Repository AI Taxonomy (Version 1). arXiv. https://doi.org/10.48550/ARXIV.2411.08054

The GREI Data Repository AI Taxonomy

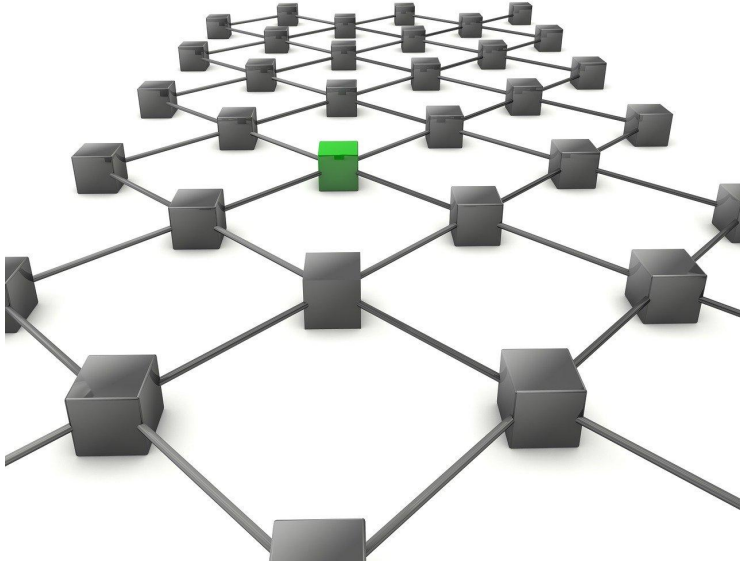


Category	Description	Example
Acquire	Gather and ingest data and metadata from different sources	AI-assisted ingest of online research data
Validate	Ensure data and metadata quality, accuracy, and integrity	AI-assisted ID of data anomalies or errors
Organize	Categorize and structure data and metadata	Tag data with keywords and descriptors
Enhance	Enrich and augment data and metadata with annotations or standard formats	Build cross-ontology knowledge graphs
Analyze	Use AI-informed analysis to uncover insights, patterns, and trends in data and metadata	AI-assisted video to text transcriptions
Share	Support data and metadata discovery, access, and distribution	AI-assisted citation generation
Support	Provide suggestions and answer questions for data and metadata users	Real-time data submission feedback

Dataverse Model Context Protocol (MCP) Server

👋**Improves**: Data findability, Data reusability, Interoperability
🤝**Who Benefits?** Developers, System integrators

- "Open standard, open-source framework standardizing how AI systems integrate and share data with external tools, systems, and data sources"
- Dataverse MCP Server: mcp.dataverse.org
- GitHub repo: https://github.com/gdcc/mcp-dataverse



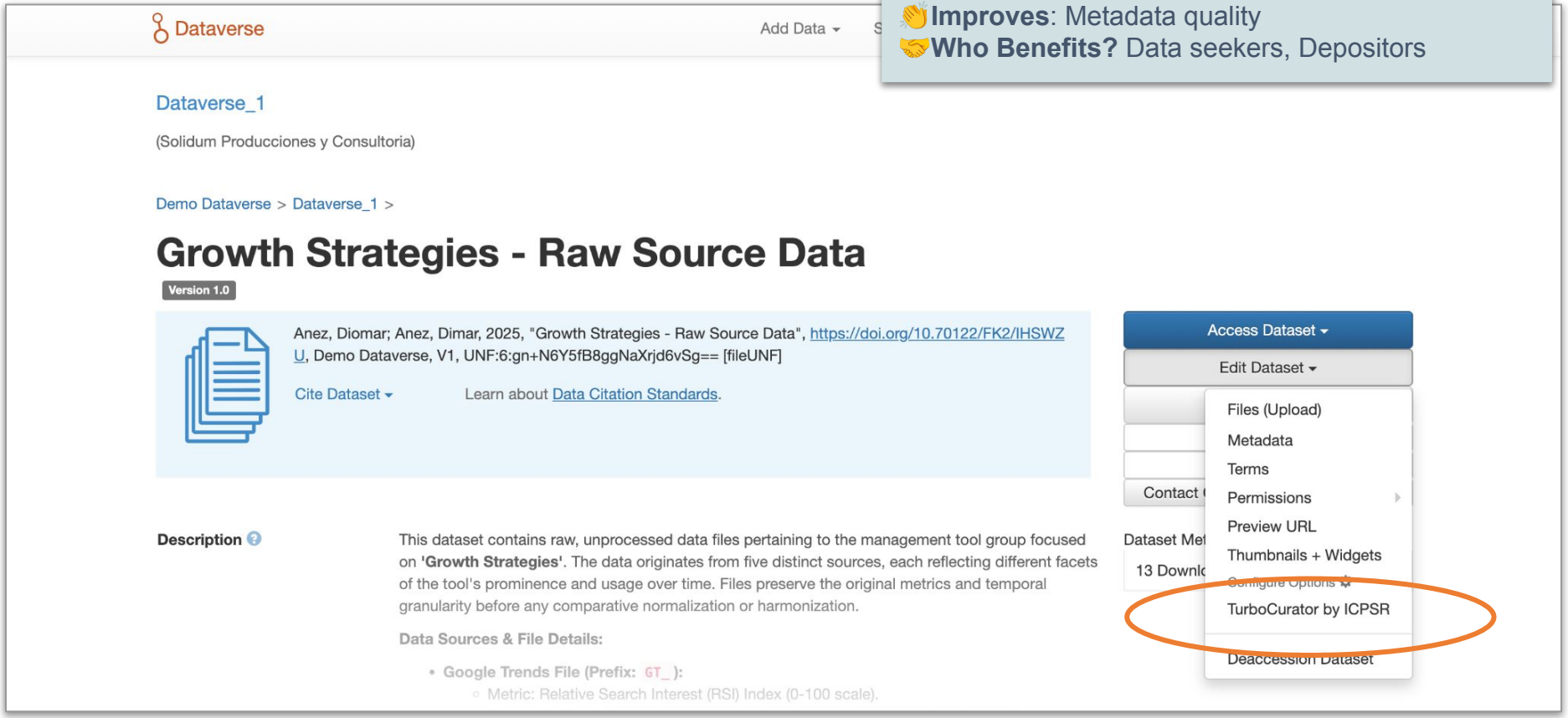
Peggy und Marco Lachmann-Anke from Pixabay

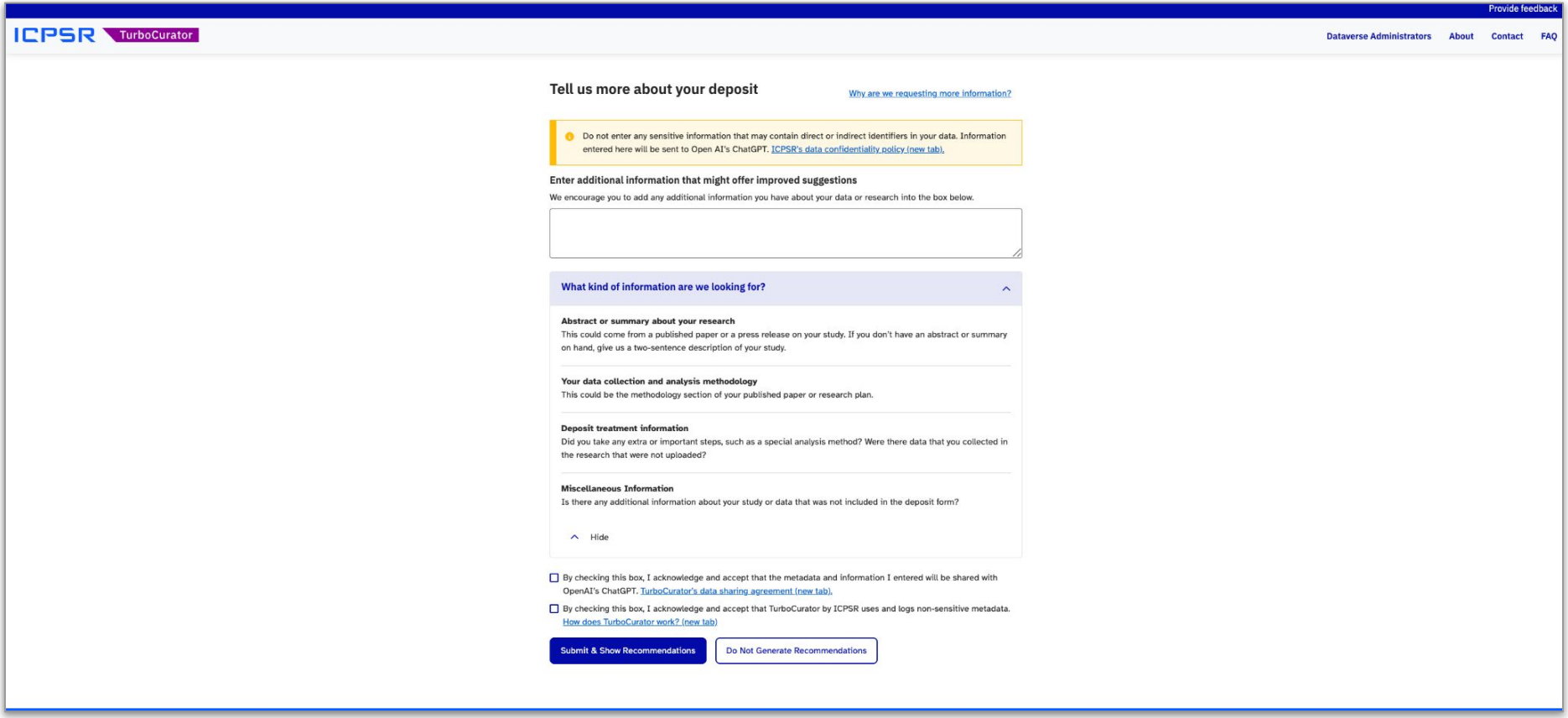
ICPSR TurboCurator

👏**Improves**: Metadata quality
🤝**Who Benefits?** Data seekers, Depositors

- TurboCurator (link)
- For Dataverse 6.1+
- Implemented as a Dataverse external tool
- Improves draft and published dataset metadata
- Uses OpenAI ChatGPT to suggest metadata fields including title, description & keywords
- Leverages the ICPSR social science thesaurus preferred terms
- For details, see the use case: TurboCurator Use Case

👏**Improves**: Metadata quality
🤝**Who Benefits?** Data seekers, Depositors




Provide feedback
ICPSR TurboCurator
Dataverse Administrators
About
Contact
FAQ
Tell us more about your deposit
Why are we requesting more information?
Do not enter any sensitive information that may contain direct or indirect identifiers in your data. Information entered here will be sent to Open AI's ChatGPT. ICPSR's data confidentiality policy (new tab).
Enter additional information that might offer improved suggestions
We encourage you to add any additional information you have about your data or research into the box below.
What kind of information are we looking for?
Abstract or summary about your research
This could come from a published paper or a press release on your study. If you don't have an abstract or summary on hand, give us a two-sentence description of your study.
Your data collection and analysis methodology
This could be the methodology section of your published paper or research plan.
Deposit treatment information
Did you take any extra or important steps, such as a special analysis method? Were there data that you collected in the research that were not uploaded?
Miscellaneous Information
Is there any additional information about your study or data that was not included in the deposit form?
Hide
By checking this box, I acknowledge and accept that the metadata and information I entered will be shared with OpenAI's ChatGPT. TurboCurator's data sharing agreement (new tab).
By checking this box, I acknowledge and accept that TurboCurator by ICPSR uses and logs non-sensitive metadata. How does TurboCurator work? (new tab)
Submit & Show Recommendations
Do Not Generate Recommendations

Current Harvard/IQSS AI Projects

	Enhancing Dataset Metadata	Research Data Metadata Knowledge Graph	Spam Detection & Workflow Automation
Goal	Extract granular, variable-level metadata at scale	Develop knowledge graph of metadata and relationships for new data acquisitions	Investigate AI technologies to reduce spam and deposits missing key information
Improves	Metadata quality	Data findability & reusability	Data curation
Who Benefits?	Data depositors Data seekers Data reusers	Data seekers Data reusers	Data curators Repository managers
Status	In development	In development	Planned

Summary: Who Benefits from Dataverse 🤖 AI?

Guides

Chatbots

Knowledge Graph

Metadata Tools

Metadata Standards

Protocols

Data curators

Data depositors

Data seekers

Dataverse repository users

Data reusers

Metadata harvesters

Repository managers

Repository strategists

Software developers

System integrators



Image by Gerd Altmann at Pixabay

Obrigado!

